YAP MING XUAN

Student Developer | AI Engineering | Computer Vision | Agentic Systems

Singapore | 221541b@student.hci.edu.sg | linkedin.com/in/russellmx | github.com/Russell-hci | https://russell-hci.github.io

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JC1 student developer and AI enthusiast with experience building applied computer vision systems, full-stack web applications, and adversarial ML research workflows. Interested in AI engineering, startup product development, agentic systems, and turning ambiguous requirements into practical prototypes.

EDUCATION

Hwa Chong Institution (Junior College) | Current JC1 Student — 2026 - 2027

- A1 in O-Level Computing; Section Head, Machine Learning Division, Infocomm & Robotics Society.

TECHNICAL SKILLS

Languages	Python, JavaScript, HTML, CSS
AI / ML	PyTorch, YOLOv8, OpenCV, Computer Vision, Deep Learning, Adversarial ML, Grad-CAM
Web / Data	React, Firebase Authentication/Database, Bootstrap, Responsive UI, Git, GitHub

SELECTED PROJECTS

MalGuard-X: Family-Balanced Adversarial Malware Classification | Python, PyTorch, EfficientNet, MobileNetV3 — CenTaD Research

- Built a malware image classification research workflow focused on adversarial robustness, combining balanced sampling, Balanced Softmax Loss, adversarial training, and robust checkpoint selection in FB-MalAT.
- Evaluated EfficientNet and MobileNetV3 under FGSM, PGD-20, and PGD-50 attacks; achieved over 80% accuracy and macro-F1 in reported adversarial evaluation settings.
- Used Grad-CAM explainability to inspect model behavior under clean and adversarial conditions.

Project SPACE: Smart Parking Accessibility Checker Enforcer | Python, YOLOv8, Roboflow, Google Colab — Computer Vision

- Developed and tested an AI-powered accessible-parking monitoring workflow using three YOLO models for vehicle detection, mobility-label detection, and license-plate detection.
- Used Roboflow for dataset annotation/versioning and Google Colab for model training; contributed to image collection, application testing, and license-plate model training.

Project SeatingPlan | JavaScript, React, Bootstrap, Firebase — Full-Stack Web App

- Built seating-plan generator functionality for teachers, supporting customizable layouts, drag-and-drop arrangement, class-list workflows, and real-time Firebase-backed retrieval/storage.
- Integrated user-facing interactions around Google/Firebase authentication, recent design access, and editing flows; project received positive teacher feedback during testing.

Pawnify Engine | JavaScript, Bootstrap, Chess.js, Chessboard.js, Stockfish.js — Project Leader

- Led a browser-based chess GUI that exposed engine customization through a no-install interface, including FEN/PGN handling, clocks, engine lines, evaluation, takebacks, and custom starting positions.
- Implemented and debugged browser-to-engine communication using Stockfish.js messaging while balancing advanced configuration with approachable UI explanations.

LEADERSHIP & EXPERIENCE

Machine Learning Section Head | Hwa Chong Institution — Infocomm & Robotics Society

- Lead ML division activities, guide members in AI/software projects, and support technical sharing sessions.

Volunteer Instructor — FutureTech Community Service Project

- Taught programming, technology, and digital-literacy concepts to Primary 5 to Secondary 2 students.

Retail Assistant — TWG Tea

- Built customer-facing communication, teamwork, and fast-paced retail operations experience.

ACHIEVEMENTS & CERTIFICATIONS

19th Place, Cyberthon 2026 | 23rd Place, BrainHack TIL-AI 2026 (Novice) | Gold Award, QCEC 2025 | Outstanding Student Award
Microsoft AI & ML Engineering Professional Certificate (Coursera) | Financial Technology Innovations Specialization (Coursera)